

Weekly Diary – Week 1

Internship Organization: Vikaspedia

Duration: 16/01/2026 – 23/01/2026

Objective of the Week

The primary objective of Week 1 was to understand the requirements of Task, set up the frontend development environment using React and Material UI, study browser-based Text-to-Speech (TTS) APIs, and implement the initial TTS features that would later support synchronized text highlighting.

Day 1

- Thoroughly reviewed the Task documentation, including its objectives, problem definition, functional expectations, and evaluation parameters.
- Understood the significance of Text-to-Speech combined with text highlighting in improving accessibility for users with low literacy levels and visual challenges.
- Analyzed project constraints such as frontend-only development, reliance on browser-supported TTS APIs, and compulsory support for Indian languages.
- Prepared a basic implementation roadmap and tentative schedule for completing the task.

Day 2

- Created a new React project following a modern, hooks-based development approach.
- Installed and configured Material UI (MUI) dependencies required for interface design.
- Designed a simple and user-friendly layout using MUI components such as Container, Typography, and Box.
- Considered responsive design principles from the initial development stage.

Day 3

- Studied the Web Speech API with specific emphasis on SpeechSynthesis and SpeechSynthesisUtterance.
- Implemented basic Text-to-Speech functionality for static text content.
- Tested different voices available across multiple browsers.
- Noted variations in language availability and voice quality among browsers.

Day 4

- Developed TTS control features including Play, Pause, Resume, and Stop.
- Managed speech playback states using React state management and effect hooks.
- Addressed edge cases such as interrupting speech and restarting playback smoothly.
- Conducted repeated tests to ensure consistent and stable TTS behavior.

Day 5

- Implemented a language selection feature using MUI dropdown components.

- Tested required Indian languages including Hindi, Marathi, Tamil, Telugu, and Gujarati.
- Added fallback handling for cases where a selected language voice was not supported by the browser.
- Recorded browser-specific limitations related to language support for future documentation.

Day 6

- Integrated speech rate control functionality using the MUI Slider component.
- Tested the impact of different speech rates on audio clarity and playback stability.
- Ensured that modifying the speech rate did not interrupt or break ongoing speech.
- Performed basic responsiveness checks on both desktop and mobile screen sizes.

Conclusion

- Successfully developed a functional React application integrated with Material UI.
- Implemented and validated core Text-to-Speech features.
- Confirmed proper working of language selection and speech rate controls.
- Established a solid foundation for implementing synchronized text highlighting in subsequent phases of the project.